

6(a)	<u>carbon dioxide</u> ;	1
6(b)	<i>any one from:</i> 1 triose phosphate dehydrogenated ; 2 <u>reduced</u> NAD, made / released ;	1
6(c)(i)	(<i>glycolysis</i>) + (<i>Krebs</i>) (<i>ATP + reduced NAD</i>) + (<i>ATP + reduced NAD + reduced FAD</i>) (4 – 2) + (4 × 3) + 2 + (6 × 3) + (2 × 2) OR 2 + 12 + 2 + 18 + 4 ; 38 ;	2

Question	Answer	Marks
6(c)(ii)	any two from: 1 ATP / energy, used to transport, pyruvate / reduced NAD / products of glycolysis, into (named part of) mitochondria ; 2 some protons leak from intermembrane space ; 3 some energy lost as heat ; 4 glucose may not be completely broken down / some intermediates are used in different metabolic processes ; 5 reduced NAD may be used for other (metabolic) reactions ;	2
6(d)	any three from: 1 coenzymes ; 2 help / for, dehydrogenases / dehydrogenation (reactions) ; 3 <i>ref. to</i> glycolysis / link reaction / Krebs cycle ; 4 carry / transfer / transport / bring, hydrogens / H ; 5 to, ETC / inner mitochondrial membrane / crista(e) ;	3

Question	Answer	Marks
6(e)	any three from: 1 less / no, energy (release from electron transfer / ETC) ; 2 less / no, chemiosmosis ; 3 fewer / no, protons, move / pumped, to intermembrane space ; 4 less (steep) / no, proton gradient ; 5 fewer / no, protons, move / diffuse, through ATP synth(et)ase ; 6 less / no, ATP synthesised ;	3

Question	Answer	Marks
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